Opencast Core Factor

1. Economic

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Economic	Valuation	Net Present Value (NPV)	NPV = Σ CFt / (1+r)^t − $CAPEX_0$	Discounted cash flow using project cash flows	Use when annual revenue, OPEX, CAPEX, and discount rate are available	Use peer multiple / benchmark valuation
Economic	Valuation	Internal Rate of Return (IRR)	Solve for r where NPV = 0	Root-finding on cash flow series	Use when cash flows are stable and multi-year forecast exists	Use NPV only
Economic	Payback	Payback Period (PBP)	Cumulative cash flow turns positive at year t	Simple cash recovery analysis	Use when investor wants capital recovery timing	Use discounted payback if cash flow is volatile
Economic	Capital	Initial $CAPEX_0$	Civil + mine development + HEMM + EPCM + contingency	Bottom-up cost build-up	Use when project setup cost is known	Use comparable mine CAPEX per tonne capacity
Economic	Capital	Sustaining CAPEX	Replacement + overhauls + expansion + infra upkeep	Annual sustaining capital schedule	Use for running mine with life-of-mine plan	Use 5–10% of initial CAPEX per year
Economic	Cost	Operating Expenditure (OPEX)	Mining OPEX + processing OPEX + admin + power + maintenance	Activity-based costing	Use when detailed operating cost data exists	Use per-tonne benchmark from comparable mines

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Economic	Cost	OB Mining Cost	OB cost per BCM removed	OB stripping cost model	Use when overburden removal is material	Use local benchmark ₹/BCM
Economic	Revenue	Coal Sale Price (CIL Notified)	Realized price per tonne	Grade-linked price realization	Use where notified price / realized price is known	Use weighted average benchmark price
Economic	Revenue	Annual Revenue	Production × Sale Price	Revenue forecast	Use when annual despatch volume is known	Use short-term average realization
Economic	Tax	Royalty Rate	Royalty = Rate × Revenue	Statutory royalty application	Use where royalty schedule is stable	Use current statutory rate as proxy
Economic	Tax	Corporate Income Tax (CIT) Rate	Tax = CIT × taxable profit	Income tax modeling	Use when taxable profit is forecastable	Use statutory tax rate only
Economic	Stripping	Break-Even Stripping Ratio (BESR)	BESR = net margin per tonne / OB cost per BCM	Economic cut-off for OC mining	Use to judge OC viability	Use historical OSR threshold of similar mines
Economic	Capital cost	EPCM Cost	EPCM = % of project cost or direct fee	Project management cost build-up	Use during project planning / expansion	Use standard EPCM percentage
Economic	Capital cost	Contingency	Contingency = % of base CAPEX	Risk buffer on capital cost	Use when design is not fully frozen	Use 10–20% depending on maturity
Economic	Finance	WACC	WACC = (E/V)Re + (D/V)Rd(1−T)	Weighted average cost of capital	Use when debt/equity structure is known	Use build-up discount rate
Economic	Finance	Discount Rate Used	Project discount rate	DCF discounting	Use when valuation is being done	Use WACC or sector benchmark

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Economic	Accounting	Annual Depreciation	CAPEX depreciated over useful life	Straight-line / tax-based depreciation	Use for profitability and tax analysis	Use book-value depreciation benchmark
Economic	Liability	Rehabilitation bond provision	Closure / rehab provision over project life	Present value of closure liability	Use when closure plan exists	Use % of CAPEX as placeholder

2.Technical

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Technical	Geometry	Instantaneous Stripping Ratio (ISR)	ISR = OB removed / coal extracted	Short-term production ratio	Use for bench / monthly planning	Use OSR if ISR unavailable
Technical	Geometry	Overall Stripping Ratio (OSR)	OSR = Total OB / Total coal	Life-to-date mine ratio	Use for long-term OC viability	Use weighted average ISR
Technical	Equipment	Shovel Bucket Capacity	m^3 per bucket	Fleet sizing / productivity model	Use when shovel specification is known	Use OEM spec
Technical	Equipment	Bucket Fill Factor	Effective fill / rated bucket	Loading efficiency estimate	Use when material fragmentation is known	Use benchmark fill factor
Technical	Equipment	Shovel Swing Cycle Time	Time per loading cycle	Time-motion analysis	Use when cycle observations exist	Use OEM cycle benchmark
Technical	Equipment	HEMM Mechanical Availability	(Scheduled − downtime) / Scheduled	Reliability analysis	Use when maintenance records exist	Use OEM availability assumption

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Technical	Equipment	HEMM Utilisation	Operating hours / Scheduled hours	Dispatch and shift analysis	Use when production logs exist	Use target utilisation benchmark
Technical	Equipment	Shovel Output Rate	Capacity × fill × cycles × availability	Productivity model	Use when shovel cycle data exists	Use fleet benchmark output
Technical	Equipment	Number of Shovels	Required output / shovel output	Fleet matching	Use for production planning	Use phased equipment count
Technical	Haulage	Dumper Payload Capacity	Tonnes per trip	Fleet capacity model	Use when truck class known	Use OEM payload
Technical	Haulage	Haul Distance (One Way)	Site haul route distance	GIS / mine plan measurement	Use when road layout is fixed	Use average routing distance
Technical	Haulage	Dumper Laden Speed	km/hr	Haul cycle model	Use when road gradient and conditions known	Use benchmark speed
Technical	Haulage	Dumper Empty Speed	km/hr	Return cycle model	Use when haul road conditions known	Use benchmark empty speed
Technical	Haulage	Truck Cycle Time	Load + haul + dump + return time	Cycle-time calculation	Use when dispatch timing is available	Use standard cycle time
Technical	Haulage	Number of Dumpers	Required haul tonnes / dumper output	Fleet sizing	Use with annual output target	Use benchmark truck-shovel ratio
Technical	Mine design	Bench Height — OB	Overburden bench height	Design geometry	Use when bench layout is fixed	Use mine design norm
Technical	Mine design	Bench Height — Coal	Coal bench height	Seam extraction design	Use when seam thickness allows	Use seam-based design height

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Technical	Drill & blast	Blast Hole Diameter	Hole dia. in mm	Blast design	Use when drilling pattern is set	Use standard hole diameter
Technical	Drill & blast	Blast Burden (B)	Distance between hole row and face	Blast geometry	Use in blast planning	Use rule-of-thumb burden
Technical	Drill & blast	Blast Spacing (S)	Distance between holes	Blast geometry	Use in blast design	Use burden-based spacing
Technical	Drill & blast	Powder Factor (PF)	kg explosive / BCM	Blast efficiency metric	Use when fragmentation target known	Use benchmark PF
Technical	Geotech	Factor of Safety — Slope (FoS)	Resisting / driving forces	Slope stability analysis	Use when geotech data is available	Use conservative FoS = 1.2–1.3
Technical	Processing	Metallurgical Recovery	Product metal / feed metal	Plant recovery estimation	Use when washery / plant data exists	Use industry recovery benchmark
Technical	Mining loss	Mining Dilution (OC)	Dilution tonnes / mined tonnes	Grade control dilution model	Use when ore-loss data exists	Use standard dilution assumption
Technical	Haul road	Haul Road Width	Road width in metres	Road design standard	Use when truck size is known	Use DGMS / OEM guideline width
Technical	Haul road	Haul Road Max Gradient (Laden)	Road slope %	Haul road design	Use for truck operability check	Use standard safe gradient
Technical	Haul road	Rolling Resistance	kg/tonne	Haulage energy model	Use when road surface condition known	Use benchmark rolling resistance
Technical	Dewatering	Pump Capacity — Installed	Total pump discharge	Water handling model	Use when pump fleet is fixed	Use capacity benchmark

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Technical	Dewatering	Total Pumping Head (H)	Static + friction head	Pump sizing	Use when groundwater inflow is known	Use design head estimate
Technical	Production	Annual production rate (MTY)	Planned tonnes / year	Mine plan output	Use when annual despatch is known	Use current dispatch average
Technical	Life	Mine life	Remaining reserve / annual production	Life-of-mine estimate	Use for operating mine valuation	Use reserve-base d estimate for this

3.Geological

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Geological	Mineable Coal Reserve	Coal seam deposits	Reserve = Area × Avg Thickness × Density × Recovery Factor	Seam correlation & isopach mapping	Borehole density ≥12 BH/km²; continuous seam laterally traceable	Interpolate between boreholes using kriging; flag as Inferred resource only
Geological	Borehole Density	Coal seam deposits	BH density (BH/km²)	Borehole spacing analysis	Use when BH density ≥12 BH/km²	Flag as Inferred resource only
Geological	Seam Thickness	Coal seam deposits	Average Thickness (m)	Borehole logging & seam correlation	Use when seam is continuous laterally	Interpolate via kriging
Geological	GCV	Resource estimation	Gross Calorific Value (kcal/kg)	Ordinary Kriging (OK)	Variogram shows spatial continuity; nugget effect <30% of sill	Weighted average of borehole intercepts
Geological	Ash Content	Resource estimation	% Ash	Ordinary Kriging (OK)	Variogram shows spatial continuity;	Weighted average of

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
					nugget effect <30% of sill	borehole intercepts
Geological	OB Thickness	Geometry	Overburden thickness (m)	Isopach mapping	Use when borehole data is sufficient	Interpolate between boreholes
Geological	Hydraulic Conductivity	Mine water balance & dewatering	m/day	Pump tests	Hydrogeological investigation complete	CGWB regional groundwater data as first approximation
Geological	UCS	Rock mass properties	Unconfined Compressive Strength (MPa)	Lab testing	Use when measured UCS is available	Fill using default values per rock type from internal database
Geological	RQD	Underground ground control risk	Rock Quality Designation (%)	Geotechnical logging of drill core	Sufficient drill core for geotechnical logging	Conservative RQD <40; use maximum support density
Geological	Mine Catchment Area	Mine water balance & dewatering	ha	GIS / Site survey	Use when topography and site layout known	Conservative 100% catchment runoff assumption

4.Environmental

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Environmental	Land	Mine Lease Area — Total	Total leased hectares	Land footprint	Use when lease boundary known	Use approved lease area

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Environmental	Land	Forest Area within Lease	Forest hectares inside lease	Land-use assessment	Use when forest demarcation exists	Use official forest records
Environmental	Dump	OB Dump Height — Per Lift	Height per lift in metres	Dump design	Use for dump stability	Use design norm
Environmental	Stability	OB Dump FoS (Bishop Simplified)	Factor of safety	Stability analysis	Use when dump geotech data exists	Use conservative FoS threshold
Environmental	Reclamation	Progressive Backfilling Ratio	Backfilled OB / total OB	Reclamation progress metric	Use for running mine	Use benchmark backfilling ratio
Environmental	Emissions	GHG Scope 1 — Direct Emissions	Fuel use × emission factor	GHG inventory	Use when diesel / fuel data exists	Use Tier 1 default emission factor
Environmental	Emissions	GHG Scope 2 — Indirect (Grid)	Grid power × emission factor	GHG inventory	Use when electricity data exists	Use regional grid factor
Environmental	Emissions	GHG Intensity	tCO_2e / tonne ROM	Emissions intensity	Use for investor ESG view	Use sector benchmark intensity
Environmental	Energy	Annual Diesel Consumption	Litres per year	Fuel accounting	Use in emissions and cost model	Use fleet benchmark fuel use
Environmental	Climate	Annual Rainfall — IMD District	mm/year	Climate loading	Use for water and flood planning	Use nearest IMD station
Environmental	Climate	Catchment Runoff Coefficient	Dimensionless runoff factor	Hydrology model	Use when catchment is known	Use conservative runoff coefficient

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Environmental	Water	Total Mine Water Inflow Rate (Q)	m^3/hr	Dewatering design	Use if hydrogeology is known	Use regional inflow estimate
Environmental	Closure	Mine closure plan status	Approved / draft / not prepared	Regulatory check	Use when closure plan exists	Use provision estimate
Environmental	Forest	Forest Clearance status	Granted / pending / refused / not required	Compliance status	Use for forest land cases	Use compliance risk flag
Environmental	Forest	Forest area pending clearance	Pending hectares	Clearance workload	Use when forest diversion exists	Use pending clearance estimate
Environmental	EC	Environmental Clearance status	Granted / conditions / pending / refused	Approval status	Use for project approval tracking	Use process stage flag

5. Social

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Social	Safety	LTIFR	$(LTI \times 10^6)$ / man-hours	Safety performance metric	Use when annual safety records exist	Use sector benchmark LTIFR
Social	Safety	FAR	$(Fatalities \times 10^8)$ / man-hours	Fatal risk indicator	Use when fatality data exists	Use benchmark FAR
Social	Workforce	Total Man-Hours Worked — Annual	Total hours worked per year	Workforce database	Use for normalization	Use payroll / shift records
Social	Safety	LTI Count — Annual	Number of lost-time injuries	Incident tracking	Use when audited incident records exist	Use industry average

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Social	Safety	Fatalities — Annual	Number of fatalities	Mandatory reporting	Use for risk scoring	Use zero if no incidents
Social	Community	Local Employment Ratio	Local workforce / total workforce × 100	Social benefit metric	Use where local hiring data exists	Use survey estimate
Social	Workforce	Total Workforce Headcount	Number of employees	HR dataset	Use for productivity / social model	Use payroll headcount
Social	Land	Project-Affected Families (PAF Count)	Number of affected families	SIA / land acquisition data	Use when land acquisition exists	Use acquisition register
Social	Land	Land Acquisition Area	Hectares acquired	R&R planning	Use when land acquisition is required	Use lease / acquisition records
Social	R&R	R&R Cost — RFCTLARR 2013	PAFs × entitlement package	Statutory rehab cost	Use when displacement exists	Use statutory package estimate
Social	CSR	CSR Mandatory Spend — 2% Net Profit	2% × average net profit	Legal CSR obligation	Use for Indian companies meeting threshold	Use voluntary CSR / SROI
Social	Capacity	Safety training hours per worker	Total training hours / workforce	Training compliance	Use when records exist	Use prescribed minimum hours

6. Risk

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Risk	Geotech	FoS — Slope Stability (Mean)	Mean factor of safety	Slope risk model	Use when slope monitoring exists	Use conservative FoS
Risk	Geotech	Std Dev of FoS (σ_FoS)	Dispersion in FoS	Monte Carlo / uncertainty model	Use when multiple slope runs exist	Use assumed variability
Risk	Geotech	Probability of Failure (POF)	POF = Φ(−β)	Reliability analysis	Use when FoS distribution exists	Use qualitative risk flag
Risk	Water	100-Year Return Flood Inflow (Q_{100})	Peak flood inflow	Flood risk model	Use for water hazard evaluation	Use conservative flood allowance
Risk	Financial	Expected Loss E[L] — Actuarial	Σ(Pi × Si)	Risk-cost model	Use when historical risk data exists	Use national average severity
Risk	Thermal	Crossing Point Temperature (CPT)	Ignition threshold temperature	Coal fire risk indicator	Use for self-heating risk	Use benchmark CPT
Risk	Gas	Seam Methane Gas Content	m^3/tonne	Gas hazard assessment	Use for coal seam risk	Use regional gas classification
Risk	Legal	Mining Lease — Years Remaining	Remaining lease tenure	Title / permit risk	Use for investment security	Use renewal-risk flag
Risk	Legal	Active Litigation Count	Number of disputes/cases	Legal due diligence	Use when legal records available	Use risk score if unknown

7. Geographical

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Geographical	Logistics	Mine-to-Rail Siding Distance	Distance in km	GIS / route mapping	Use when dispatch route known	Use nearest transport corridor
Geographical	Logistics	Rail Freight Tariff — FOIS	₹/tonne-km	Rail cost model	Use when FOIS tariff is known	Use current rail benchmark
Geographical	Logistics	Road Haulage Cost	₹/tonne-km	Road transport model	Use when trucking is used	Use local haulage benchmark
Geographical	Output	Annual Coal Despatch Volume	Mtpa	Dispatch forecasting	Use when offtake / dispatch exists	Use production average
Geographical	Logistics	Total Logistics Cost	Freight + loading + unloading + handling	Delivered cost model	Use for investor margin view	Use benchmark landed cost
Geographical	Working days	Annual Working Days	365 − monsoon − shutdowns	Operations calendar	Use when climate data exists	Use standard working days
Geographical	Climate	Monsoon Disruption Days	Days lost due to rain	Climate risk adjustment	Use for Indian mines	Use IMD seasonal average
Geographical	Power	Grid Power Availability	% hours/year	Power reliability metric	Use when grid data exists	Use assumed availability
Geographical	Power	Annual Energy Demand	Total MWh/year	Load estimation	Use for power planning	Use equipment-based estimate
Geographical	Power	Power Tariff — DISCOM	₹/kWh	Cost planning	Use when grid tariff is known	Use current tariff schedule
Geographical	Water	Process Water Demand	m^3/day	Plant / mine water requirement	Use when mine water balance exists	Use benchmark demand

Subtopic part

1.Mine Life

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Mine Life	Reserve	Mineable Reserve (ROM)	Remaining ROM reserve	Reserve depletion model	Use when reserve balance is known	Use latest reserve statement
Mine Life	Geometry	Overall Stripping Ratio (OSR)	OB / coal	Long-term mine life indicator	Use when mine is in operation	Use average OSR of last 3 years
Mine Life	Equipment	HEMM Availability (Fleet Weighted)	Weighted availability across fleet	Reliability-weighted fleet analysis	Use when equipment logs exist	Use OEM guaranteed availability
Mine Life	Equipment	HEMM Utilisation	Operating hours / scheduled hours	Dispatch efficiency	Use when shift data exists	Use benchmark utilisation

2. HEMM and Cost

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
HEMM and Cost	Shovel	Shovel Bucket Capacity	m^3	Productivity modelling	Use when shovel class known	Use OEM capacity
HEMM and Cost	Loading	Bucket Fill Factor	Dimensionless	Loading efficiency	Use when fragmentation is known	Use benchmark factor
HEMM and Cost	Loading	Shovel Swing Cycle Time	Seconds	Time-motion study	Use site measurements	Use OEM cycle
HEMM and Cost	Fleet	HEMM Mechanical Availability	%	Reliability analysis	Use maintenance records	Use benchmark availability

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
HEMM and Cost	Fleet	HEMM Utilisation	%	Dispatch analysis	Use production logs	Use benchmark utilisation
HEMM and Cost	Output	Shovel Output Rate	BCM/hr	Capacity × fill × cycle × availability	Use when fleet data exists	Use estimated output
HEMM and Cost	Fleet	No. of Shovels	Count	Output target / unit output	Use when fleet plan is known	Use matching ratio
HEMM and Cost	Haulage	Dumper Payload	Tonnes	Fleet selection	Use truck specs	Use OEM payload
HEMM and Cost	Haulage	Haul Distance (One Way)	Metres	Haul route design	Use mine layout	Use average route
HEMM and Cost	Haulage	Dumper Laden Speed	km/hr	Cycle model	Use haul road profile	Use benchmark speed
HEMM and Cost	Haulage	Dumper Empty Speed	km/hr	Return cycle model	Use haul road profile	Use benchmark speed
HEMM and Cost	Haulage	Truck Cycle Time	Minutes	Load + haul + dump + return	Use when truck cycle data exists	Use standard cycle
HEMM and Cost	Fleet	No. of Dumpers	Count	Annual output / truck capacity	Use for fleet sizing	Use industry ratio
HEMM and Cost	Capital	Total HEMM + Civil CAPEX	Total asset cost	Capex estimation	Use when equipment and civil scope are known	Use comparable mine CAPEX
HEMM and Cost	OPEX	OPEX (All-in per tonne)	Total cost / tonne	Unit cost model	Use when full cost data exists	Use benchmark OPEX
HEMM and Cost	OB Cost	OB Mining Cost	₹/BCM	OB cost model	Use for stripping cost	Use local OB benchmark

3. Stripping Ratio

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Stripping Ratio	Instantaneous	ISR	OB / coal for current slice	Short-term operational analysis	Use for bench-level planning	Use OSR if ISR unavailable
Stripping Ratio	Overall	OSR	Total OB / total coal	Mine-life stripping ratio	Use for OC feasibility	Use average production ratio
Stripping Ratio	Economic	BESR	Net coal margin / OB mining cost	Break-even economic test	Use for valuation / mine cut-off	Use adjacent mine benchmark
Stripping Ratio	Price	Coal Sale Price	₹/tonne	Revenue realization	Use when grade-wise price known	Use notified benchmark price
Stripping Ratio	Cost	Coal OPEX	₹/tonne	Operating cost analysis	Use when cost data exists	Use benchmark coal OPEX
Stripping Ratio	Cost	OB Mining Cost	₹/BCM	OB removal costing	Use when equipment cost known	Use local benchmark
Stripping Ratio	Statutory	Royalty Rate	%	Statutory deduction	Use when royalty known	Use current statutory rate
Stripping Ratio	Geometry	OB Thickness	m	Cross-section / model	Use for cut design	Use average section
Stripping Ratio	Geometry	Seam Thickness	m	Geological model	Use for mine planning	Use seam average
Stripping Ratio	Geometry	Seam Dip	degrees	Structural interpretation	Use for mining method selection	Use local seam orientation

4. Coal Quality and Washery

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Coal Quality	Heat value	GCV — Seam-wise	kcal/kg	Lab testing	Use when seam samples exist	Use seam average
Coal Quality	Heat value	Blended GCV	Weighted average GCV	Blend planning	Use when multiple seams are blended	Use production-weighted GCV
Coal Quality	Price	CIL Notified Price for Grade	₹/tonne	Grade price application	Use when grade is notified	Use price benchmark
Coal Quality	Impurity	Ash Content	%	Proximate analysis	Use when quality data exists	Use grade benchmark
Coal Quality	Moisture	Total Moisture (TM)	%	Lab analysis	Use for washery and pricing	Use average grade moisture
Coal Quality	Moisture	Inherent Moisture (IM)	%	Lab analysis	Use for combustion behavior	Use average IM
Coal Quality	Combustion	Volatile Matter (VM)	%	Proximate analysis	Use for coal classification	Use benchmark VM
Coal Quality	Combustion	Fixed Carbon (FC)	100 − ash − moisture − VM − S	Derived from proximate analysis	Use when full analysis exists	Use calculated average
Coal Quality	Emission	Sulphur Content	%	Lab analysis	Use for emissions / blending	Use grade average
Coal Quality	Safety	Crossing Point Temperature (CPT)	°C	Spontaneous combustion indicator	Use for fire risk	Use benchmark CPT

5. Bench and Design

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Bench and Design	Geometry	OB Bench Height	Metres	Pit design standard	Use when slope/bench design fixed	Use approved design height
Bench and Design	Geometry	Coal Bench Height	Metres	Seam extraction design	Use when seam thickness permits	Use seam-based bench height
Bench and Design	Blast	Blast Hole Diameter (d)	mm	Drill pattern design	Use when drilling data exists	Use standard hole dia.
Bench and Design	Blast	Blast Burden (B)	Metres	Blast geometry	Use in fragmentation design	Use rule-of-thumb burden
Bench and Design	Blast	Blast Spacing (S)	Metres	Blast geometry	Use in blast design	Use burden-based spacing
Bench and Design	Blast	Powder Factor (PF)	kg/BCM	Explosive efficiency	Use when fragmentation target known	Use benchmark powder factor
Bench and Design	Geotech	OB UCS — Rock Strength	MPa	Rock testing	Use when lab data exists	Use strength class estimate
Bench and Design	Geotech	RQD — OB Core	%	Core logging	Use for slope/blast design	Use conservative RQD
Bench and Design	Stability	Highwall FoS (Bishop)	Dimensionless	Slope stability	Use when geotech model exists	Use conservative FoS
Bench and Design	Haul Road	Haul Road Width	Metres	Road design	Use for dump/truck traffic	Use OEM/DGMS guideline
Bench and Design	Haul Road	Rolling Resistance	kg/tonne	Haulage energy model	Use when road condition known	Use benchmark resistance

6. Dewatering

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Dewatering	Hydrogeology	Hydraulic Conductivity (k)	m/day	Pump test interpretation	Use when aquifer testing exists	Use regional hydro value
Dewatering	Hydrogeology	Aquifer Thickness	metres	Hydrogeological model	Use when bore logs exist	Use estimated thickness
Dewatering	Hydrogeology	Depth Below Water Table	metres	Groundwater risk	Use for inflow planning	Use conservative depth
Dewatering	Hydrology	Mine Catchment Area	km^2	Catchment mapping	Use for inflow estimation	Use watershed estimate
Dewatering	Climate	Annual Rainfall (IMD)	mm/yr	Rainfall data	Use for seasonal inflow	Use nearest IMD station
Dewatering	Hydrology	Runoff Coefficient	Dimensionless	Hydrologic model	Use when land use is known	Use conservative runoff factor
Dewatering	Water	Total Mine Water Inflow (Q)	m^3/hr	Inflow estimation	Use when water data exists	Use regional inflow benchmark
Dewatering	Equipment	Pump Capacity — Installed	m^3/hr	Pump sizing	Use when pump list exists	Use capacity benchmark
Dewatering	Equipment	Total Pumping Head (H)	metres	Pump head model	Use for power sizing	Use conservative head
Dewatering	Cost	Power Tariff (for pumping cost)	₹/kWh	Operating cost model	Use when tariff known	Use DISCOM tariff

7. Infrastructure

Category	Sub-factor	Type	Formula	Method	Condition	Fallback
Infrastructure	Transport	Mine-to-Rail Siding Distance	km	GIS route distance	Use when dispatch corridor known	Use nearest siding distance
Infrastructure	Transport	FOIS Rail Freight Tariff	₹/tonne-km	Rail freight model	Use when rail tariff known	Use current tariff benchmark
Infrastructure	Transport	Road Haulage Cost	₹/tonne-km	Road logistics model	Use when road movement is needed	Use benchmark road cost
Infrastructure	Output	Annual Despatch Volume	Mtpa	Dispatch forecast	Use when sales/offtake exists	Use production average
Infrastructure	Cost	Total Logistics Cost	₹/tonne	Freight + handling + loading	Use for delivered-cost view	Use benchmark delivered cost
Infrastructure	Power	Grid Power Availability	% hours/year	Reliability analysis	Use when grid is connected	Use assumed reliability
Infrastructure	Power	Annual Energy Demand	MWh/yr	Load estimation	Use for mine operations	Use equipment-based estimate
Infrastructure	Power	Power Tariff — DISCOM	₹/kWh	Power cost model	Use when tariff published	Use current tariff schedule
Infrastructure	Water	Process Water Demand	m^3/day	Water balance	Use when process demand known	Use benchmark plant demand

Mining Project Scoring Model (0–100)

Step 1: Category Weights (same as your framework)

Category	Weight
Technical	25
Economic	30
Environmental	10
Social	15
Geographical	10
Risk	10
Total	**100**

Step 2: Sub-factor Scoring (Derived from your Excel parameters)

1. Economic (30%)

Sub-factor	Parameter	Scoring Logic
Profitability	NPV	>1500 Cr → 5, 500–1500 → 4, <500 → 2, N/A → 1
Profitability	IRR	>25% → 5, 15–25 → 4, <15 → 2
Cost	OPEX (₹/t)	<800 → 5, 800–1200 → 3, >1200 → 1
Revenue	Coal Price	Higher vs peers → higher score
Efficiency	BESR vs OSR	OSR < BESR → 5, ≈ → 3, > → 1
Capital	CAPEX efficiency	Lower CAPEX per MT → higher score
👉 **Final Economic Score:**		**Economic Score = (Sum of sub-scores / max possible) × 30**

2. Technical (25%)

Sub-factor	Parameter	Scoring Logic
Geometry	OSR	<2 → 5, 2–4 → 3, >4 → 1
Equipment	HEMM Availability	>85% → 5, 75–85 → 3, <75 → 1
Equipment	HEMM Utilisation	>75% → 5
Productivity	Shovel Output	Higher vs dataset → higher score
Design	FoS	>1.4 → 5, 1.3–1.4 → 3, <1.3 → 1
Production	MTY	Higher production → higher score

3. Environmental (10%)

Sub-factor	Parameter	Scoring Logic
Compliance	EC Status	Granted → 5, Conditions → 3, Pending → 1
Forest	FC Status	Not required → 5, Granted → 4, Pending → 2
Emissions	GHG Intensity	Lower → better
Reclamation	Backfilling Ratio	>60% → 5
Water	Inflow control	Stable → 5

4. Social (15%)

Sub-factor	Parameter	Scoring Logic
Safety	LTIFR	<4 → 5, 4–6 → 3, >6 → 1
Safety	FAR	<1 → 5
Workforce	Local Employment	>70% → 5
Impact	PAF Count	Lower → better
CSR	Spend	Meets mandate → 5

5. Geographical (10%)

Sub-factor	Parameter	Scoring Logic
Logistics	Distance to rail	<10 km → 5
Cost	Logistics cost	Lower → better
Climate	Working days	>320 → 5
Power	Availability	>90% → 5
Water	Availability	Adequate → 5

6. Risk (10%)

Sub-factor	Parameter	Scoring Logic
Geotech	FoS	>1.4 → 5
Failure	POF	<2% → 5
Flood	Q100	Lower → better
Legal	Litigation	0 → 5
Lease	Years remaining	>20 → 5